side state—in fact it offers many advantages over each storage strategy on its own:

- Database tokens can be easily and immediately revoked. In September 2018, Facebook was hit by an attack that exploited a vulnerability in some token-handling code to quickly gain access to the accounts of many users (https://newsroom.fb.com/news/2018/09/security-update/). In the wake of the attack, Facebook revoked 90 million tokens, forcing those users to reauthenticate. In a disaster situation, you don't want to be waiting hours for tokens to expire or suddenly finding scalability issues with your blocklist when you add 90 million new entries.
- On the other hand, plain database tokens may be vulnerable to token theft and forgery if the database is compromised, as described in section 5.3 of chapter 5. In that chapter, you hardened database tokens by using the `HmacTokenStore` to prevent forgeries. Wrapping database tokens in a JWT or other authenticated token format achieves the same protections.
- Less security-critical operations can be performed based on data in the JWT alone, avoiding a database lookup. For example, you might decide to let a user see which Natter social spaces they are a member of and how many unread messages they have in each of them without checking the revocation status of the token, but require a database check when they actually try to read one of those or post a new message.